

- 5 An electron enters a region R perpendicularly to a uniform magnetic field which is directed into the plane of the paper. The electron's initial velocity is also perpendicular to a uniform electric field which is directed downwards as shown in Fig. 5.1.

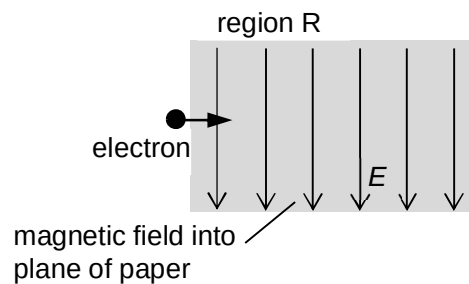
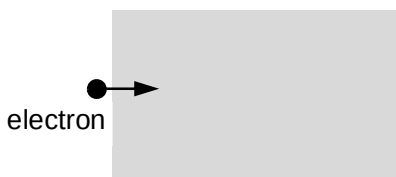


Fig. 5.1

The magnetic flux density B of the magnetic field and electric field strength E are adjusted such that the electron emerges undeviated.

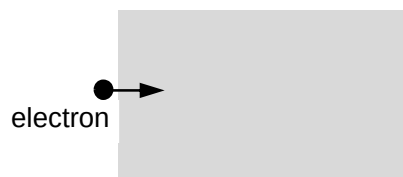
(a) Sketch and describe the path of the electron when

- (i)** only the magnetic field is turned on,



..... [1]

- (ii)** only the electric field is turned on.



..... [2]

- (b) When the electric field is turned off and a magnetic field of constant flux density B is maintained, the electron moves in a circular path of radius 1.2 cm with a speed of $1.8 \times 10^8 \text{ m s}^{-1}$.

Calculate the magnitude of the electric field strength E such that the electron emerges undeviated when the electric field is turned on.

$$E = \dots\dots\dots \text{ V m}^{-1} \text{ [3]}$$

- (c) In Fig. 5.1, the electron is replaced with a proton with a speed of $9.0 \times 10^7 \text{ m s}^{-1}$.

Describe and explain the path of the proton if both E and B remain at the same magnitude and direction as in (b).

.....

 [2]

6 (a) Explain what is meant by the term 'root-mean-square value' as applied to an